Lenticel

Lenticel is raised opening or pore on the epidermis or bark of stems and roots.

It is formed during secondary growth in stems. When phellogen is more active in the region of lenticels, a mass of loosely arranged thin-walled parenchyma cells are formed. It is called **complementary tissue** or **filling tissue**.

Lenticel is helpful in exchange of gases and transpiration called **lenticular transpiration**.

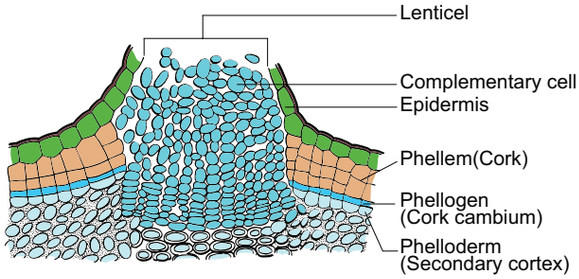


Figure 10.14: Structure of Lenticel

10.2 Secondary Growth in Dicot root

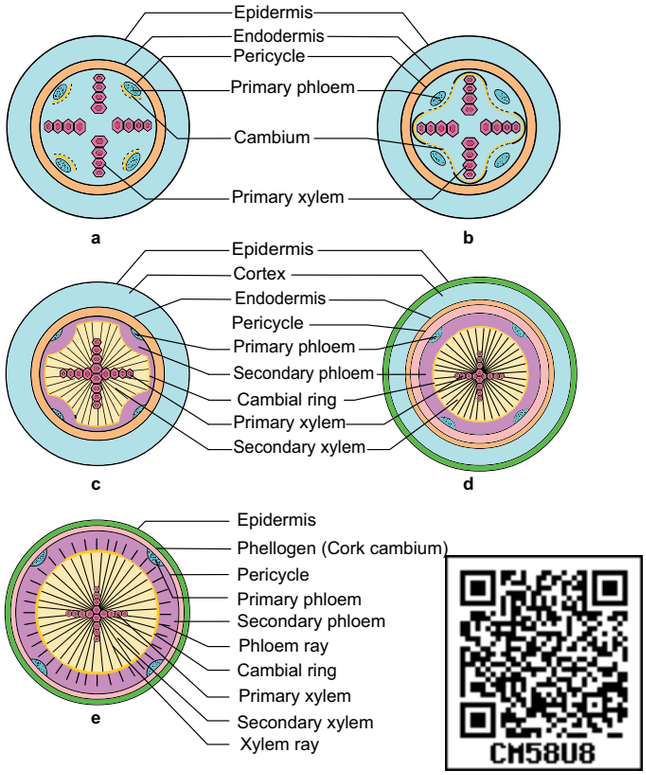


Figure 10.15: Different stages of the secondary growth (diagrammatic) in a typical dicot root (a–e)

Secondary growth in dicot roots is essential to provide strength to the growing aerial parts of the plants. It is similar to that of the secondary growth in dicot stem. However, there is marked difference in the manner of the formation of vascular cambium.

The vascular cambium is completely secondary in origin. It originates from a combination of conjunctive tissue located just below the phloem bundles, and as a portion of pericycle tissue present above the protoxylem to form a complete and continuous wavy ring. This wavy ring later becomes circular and produces secondary xylem and secondary phloem similar to the secondary growth in stems.

Differences Between Secondary Growth in Dicot Stem and Root

Secondary growth in dicot stem	Secondary growth in dicot root
The cambial ring formed is circular in cross section from the beginning.	The cambial ring formed is wavy in the beginning and later becomes circular.
The cambial ring is partially primary (fascicular cambium)and partially secondary (Interfascicular cambium) in origin.	The cambial ring is completely secondary in origin.
Generally, periderm originates from the cortical cells (extrastelar in origin).	Generally, periderm originates from the pericyle. (intrastealar in origin)
More amount of cork is produced as stem is aboveground	Generally, less amount of cork is produced as root is underground.
Lenticels of periderm are prominent.	Lenticels of periderm are not very prominent.

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